# Emanuele Donno

Computer Engineer | ML/AI Engineer | PhD Student | Quantum Computing & Quantum ML

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## PROFESSIONAL SUMMARY

Licensed **Computer Engineer** (Order of Engineers, Lecce – Sec. A, Sector C, since 06/2024) and **PhD Student** in Engineering of Complex Systems at Università del Salento, working on **Quantum Computing** and **Quantum Machine Learning** for Climate Science. About 18 months as Post-Degree Researcher at the Euro-Mediterranean Centre on Climate Change (CMCC), working on **end-to-end ML/DL pipelines** on multidimensional NetCDF climate datasets and on **ML4Fires**, the wildfire-forecasting thematic module of the EU **InterTwin** Digital Twin project, on HPC infrastructure. Full-stack engineering background from MSc projects and thesis work, covering **Python**, **Java/Spring Boot REST**, **Angular/Ionic**, **Docker**, **Git** and **MPI/HPC**. First author of a preprint on quantum data assimilation under review at FGCS; co-author of peer-reviewed and preprint contributions on Digital Twins and ML for extreme weather events.

## TECHNICAL SKILLS

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|---|---|
| **Languages** | Python, Java, TypeScript, C, SQL, Shell/Bash, Matlab, LaTeX |
| **AI & ML** | PyTorch, TensorFlow, Keras, Scikit-Learn, Computer Vision, end-to-end ML pipelines, distributed training |
| **LLM & GenAI** *(personal projects)* | Ollama (local LLMs), Hugging Face, RAG, prompt engineering, LLM-assisted dev workflow (Claude Code, ChatGPT, Gemini) |
| **Quantum** *(learning during PhD)* | Qiskit, PennyLane, Quantum Computing, Quantum Machine Learning, Quantum Circuits, Quantum Data Assimilation |
| **Backend & Infra** | Spring Boot, REST APIs, Docker, Git, MPI/MPICH, HPC, Linux/Unix |
| **Frontend** | Angular, Ionic, TypeScript, HTML5/CSS, Firebase |
| **Data & Storage** | NumPy, Pandas, NetCDF, Xarray, Cartopy, MySQL, MongoDB, SQLite |

## PROFESSIONAL EXPERIENCE

**Post-Degree Researcher** | *CMCC Foundation – ADIC Division (Lecce, Italy)* 06/2023 – 11/2024

Advanced Scientific Computing for Climate Change

- Worked on **ML4Fires** (github.com/CMCC-Foundation/ML4Fires), the wildfire-forecasting thematic module of the EU **InterTwin** project (Grant Agreement 101058386), within the Environmental Digital Twin Engine for wildfire propagation and projection at global scale.
- Worked on **end-to-end ML/DL pipelines** (data preparation, feature engineering, training, evaluation) for CNN, U-Net and U-Net++ architectures on a global multi-source climate datacube called **SeasFire Cube v3** (NetCDF/Xarray) and CMIP6 SSP projections in inference.
- Helped migrate the training stack from **TensorFlow to PyTorch** and run it in **multi-node, multi-GPU distributed training** on HPC infrastructure; ran containerized inference on HPC.
- Adopted **MLflow** for experiment tracking and reproducibility of training runs.

## EDUCATION

**PhD in Engineering of Complex Systems** 2024 – Present

*Università del Salento – Lecce, Italy*

Research project: *Quantum Computing and Quantum Machine Learning approaches for Climate Change applications*.

**M.Sc. in Computer Engineering** | Grade: 105/110 2019 – 2023
*Università del Salento – Lecce, Italy*
Thesis: *Global downscaling of remotely sensed soil-moisture retrievals using Machine Learning.*

Relevant coursework: Big Data Management, High Performance Computing, Computer Vision, Software Engineering, Internet of Things, Parallel & Sequential Algorithms, Network Technologies.

**B.Sc. in Information Engineering** | Grade: 97/110 2016 – 2019
*Università del Salento – Lecce, Italy*
Thesis: *Design and development of a back-end subsystem for an event-management service (Spring Boot REST).*

## PUBLICATIONS & PREPRINTS

Full list and metrics: orcid.org/0009-0006-2126-4816

- **interTwin: Advancing Scientific Digital Twins through AI, Federated Computing and Data.**
  *Future Generation Computer Systems (FGCS) – The International Journal of eScience*, 2026. Journal article.
  DOI: 10.1016/j.future.2025.108312 (WOS:001651938800001)
- **From Classical to Quantum-Mechanical Data Assimilation: A Comparison between DATO and QMDA.**
  E. Donno, G. Conti, P. Oddo, S. Gualdi, L. Mainetti, G. Aloisio.
  *Submitted to Future Generation Computer System (FGCS), Special Collection on Advances in QuantComputing: Methods, Algorithms, and Systems. Vol. III (Under review)*
  *Preprint (ArXiv)*, May 2026. arXiv: 2605.04881
- **Facilitating the development of Machine Learning-based Digital Twin applications for extreme weather events.**
  D. Elia, E. Donno, M. Bunino, M. Fronza, D. Donno, G. Padovani, S. Fiore, A. Manzi.
  *EGU General Assembly 2025 – Preprint*, March 2025.
  DOI: 10.5194/egusphere-egu25-13888
- **A Data Space environment for Big Data and ML-based Climate applications in the European Open Science Cloud.**
  D. Elia, F. Antonio, S. Fiore, E. Donno, G. Accarino, P. Nassisi, G. Aloisio.
  *EGU General Assembly 2024 – Preprint*, January 2025.
  DOI: 10.5194/egusphere-egu24-17472

## SELECTED PROJECTS

- **ML4Fires – InterTwin Digital Twin Engine**
  Core ML/DL component of an Environmental Digital Twin for global wildfire propagation, integrated into the EU InterTwin platform; reproducible training pipeline distributed on HPC. github.com/CMCC-Foundation/ML4Fires
  PyTorch DeepLearning DigitalTwin Climate HPC Python
- **Global Soil-Moisture Downscaling**
  ML-based downscaling of satellite-derived soil-moisture products on the EASE 2.0 grid; pipelines for multidimensional Earth-observation datasets.
  MachineLearning DeepLearning RemoteSensing NetCDF Xarray Python PyTorch
- **Social Tech Platform – Full-Stack**
  LinkedIn-style multi-user platform: Angular + Ionic frontend, Spring Boot REST backend, Firebase for

push and bulk notifications; end-to-end ownership from architecture to deployment.

Angular SpringBoot Ionic Firebase TypeScript REST

- **Drupal CMS Plugin for Remote Learning**
  Drupal content-management plugin backed by a Spring Boot service over a Materialized-Table relational schema; Angular web client; stored procedures for config/data exchange between backends.
  Drupal SpringBoot MySQL Angular Java
- **MAD – My Accessible Data**
  Multi-user system for secure personal-data management using IOTA's Masked Authenticated Messaging (MAM) protocol over the Tangle blockchain; Angular/Ionic frontend, Spring Boot REST backend.
  Blockchain IOTA IoT Java Angular
- **Parallel Fast Hartley Transform (FHT)**
  Parallel design and implementation of the Fast Hartley Transform (FHT) in the one-dimensional case, using the message passing C library MPICH.
  C MPI ParallelComputing
- **COVID-19 Spread Modeling**
  Polynomial and multivariate regression, and Kalman filtering applied to Covid-19 spread modeled with the SIR compartmental model.
  Python DataAnalysis Scipy Scikit-Learn

## CERTIFICATIONS & AWARDS

- **Samsung Innovation Camp** – Diploma in Innovation & Technology, Business Strategy, Digital Marketing (Mar. 2019).
- **IELTS Academic** – English proficiency, Overall 6.0/9.0, CEFR B2 (Jul. 2019).
- **ICIAP 2021** – 21st International Conference on Image Analysis and Processing (May 2022).
- **Professional Engineering License** – Order of Engineers of Lecce (Sec. A, Sector C, N° 4507), since 25/06/2024.
- **JUNIQ-EPIQ Summer School on Quantum Computing** – Forschungszentrum Jülich; gate-based algorithms and quantum annealing (Sep. 2025).
- **Quantum Machine Learning with Python & Qiskit** – IBM SkillsBuild; building basic and advanced models, implementing algorithms, challenges and future prospects (May 2026).
- **Introduction to FastAPI** – DataCamp (Jun. 2026).
- **Introduction to Power BI** – DataCamp (Jun. 2026).
- **Large Language Models (LLMs) Concepts** – DataCamp (Jun. 2026).

**Languages.** Italian (native), English (CEFR B2 – IELTS Academic 6.0/9.0).